

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1**

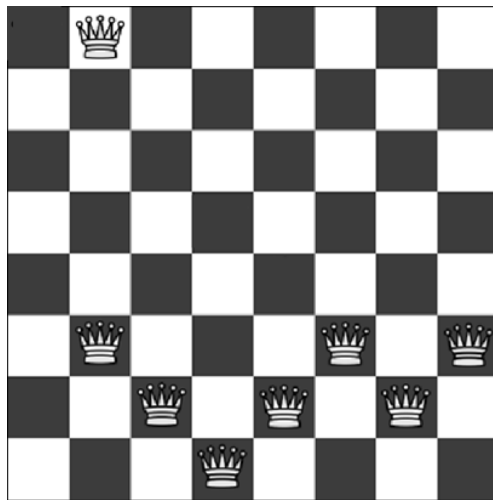
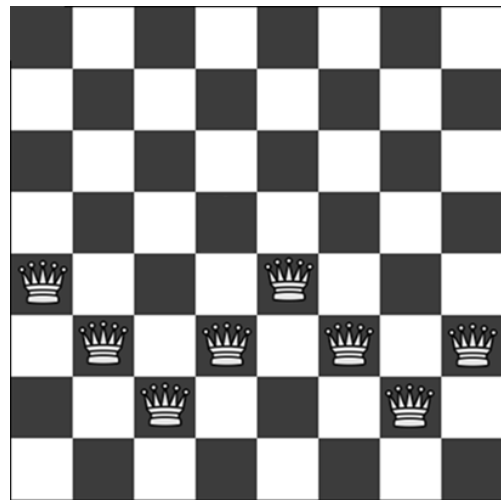
- a) The *Turing Test*, introduced by Alan Turing in 1950, is a widely recognised test in the field of Artificial Intelligence (AI). In contrast, the *Total Turing Test*, which was introduced by Stevan Harnad in 1991, serves as a more rigorous benchmark for AI.
- (i) Describe the purpose of the *Turing Test* in your own words. (1 mark)
 - (ii) Describe how the *Turing Test* works and the significance of its outcome in your own words. (3 + 1 marks)
 - (iii) Identify **TWO (2)** additional requirements of the *Total Turing Test* compared to the original *Turing Test*. (2 marks)
 - (iv) Describe the **SIX (6)** capabilities a computer must possess to pass the *Total Turing Test*. (6 marks)
- b) In AI, an environment can be thought of as the world in which an intelligent entity, usually known as agent, operates and makes decisions. Since the range of environments in AI is vast, they are categorised into the following six dimensions to guide the design of agents as well as determining the applicability agent implementation:
- Single agent vs. multiagent
 - Fully observable vs. partially observable
 - Deterministic vs. non-deterministic
 - Episodic vs. sequential
 - Static vs. dynamic
 - Discrete vs. continuous

Google Photos, a *facial recognition application*, can automatically group photos based on the people in them, and help users organise and search their photo library. Suppose its environment is images. Identify the properties of the environment based on the **SIX (6)** dimensions mentioned earlier and justify your answer. (6 + 6 marks)

[Total: 25 marks]

Question 2

- a) In AI, *classical search* and *local search* are two common strategies for solving search problems, with the latter often used for optimisation.
- (i) Provide **TWO (2)** examples for each type of search strategy. (4 marks)
 - (ii) Differentiate between these two types of strategies in **TWO (2)** distinct ways. (4 marks)
- b) *Hill climbing* is a popular local search technique, especially useful for optimisation problems.
- (i) Explain what *hill climbing* is in your own words. (3 marks)
 - (ii) Explain **THREE (3)** advantages of using *hill climbing*. (3 marks)
 - (iii) Propose **ONE (1)** *heuristic cost function* and **ONE (1)** *heuristic objective function* for applying *hill climbing* to solve the *8-Queens problem*. Then, illustrate a *state-space landscape* for each function, where the elevation corresponds to the respective function. (**Remark:** You must label the current state, shoulder, local minimum/maximum, global minimum/maximum, as well as the x-axis and y-axis correctly.) (2 + 7 marks)
 - (iv) Based on the two *8-Queens problem* states provided in **Figure 1** and **Figure 2**, compute their heuristic values using the *heuristic cost function* proposed in the **Question 2 b) (iii)**. (2 marks)

**Figure 1.** State x of 8-Queen problem**Figure 2.** State y of 8-Queen problem

[Total: 25 marks]

Question 3

Adversarial search aims to find optimal strategies in environments where multiple agents compete. It is widely used in competitive games like chess and tic-tac-toe. Two common adversarial strategies are *minimax search* and *alpha-beta pruning*.

- a) Explain what *minimax search* is in your own words. (2 marks)
- b) Explain what *alpha-beta pruning* is in your own words. (2 marks)
- c) Answer the following questions based on a simple *game tree* provided in **Figure 3**.

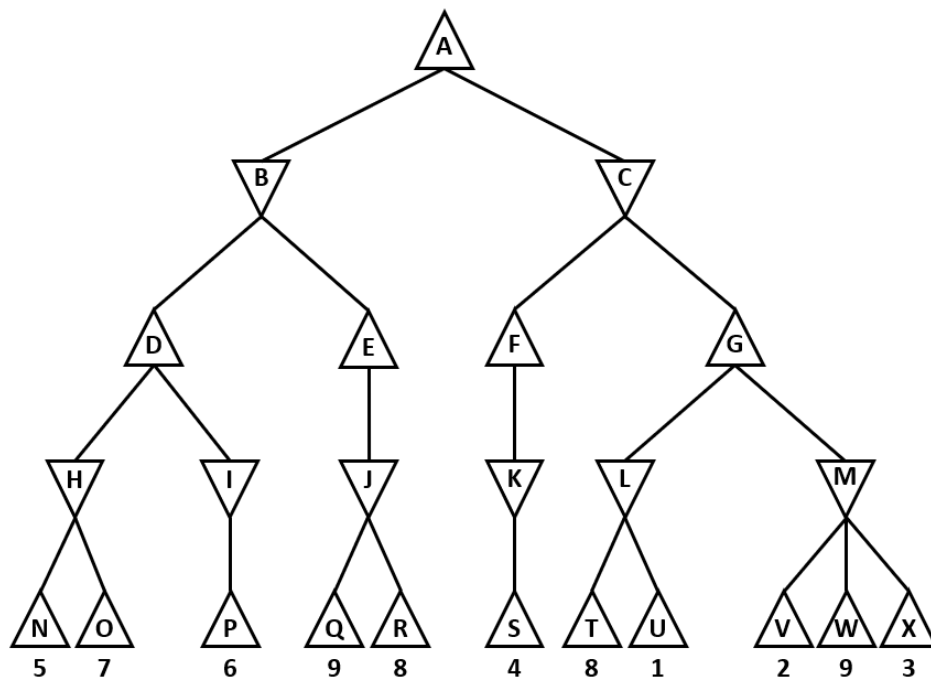


Figure 3. A game tree

- (i) Compute the *minimax* value for 12 nodes from A to M. (6 marks)
- (ii) Identify the best move for the *MAX* player at the root node. (1 mark)
- (iii) Apply the *minimax search* with *alpha-beta pruning* in *left-to-right* order and identify the nodes that would be pruned. Then, apply the *minimax search* with *alpha-beta pruning* in *right-to-left* order and identify the nodes that would be pruned. (**Remark:** For each order, you must demonstrate how alpha-beta pruning works on the game tree provided in **Figure 3**.) (13 marks)
- (iv) Derive a conclusion based on your answers to **Question 3 c) (iii)**. (1 mark)

[Total: 25 marks]

Question 4

Sudoku is a puzzle game that involves the logical placement of numbers and has introduced millions to the concept of *Constraint Satisfaction Problems (CSP)*. **Figure 4** shows an example of a 4×4 Sudoku grid consisting of 16 squares, some of which are pre-filled with digits between 1 and 4. The objective is to fill the grid so that each row, column, and 2×2 sub-grid contains the digits 1 through 4. Suppose the *Alldiff* global constraint is applied to solve the Sudoku in **Figure 4**. Answer the following questions:

	1	2	3	4
A	1			4
B			3	
C	3			2
D		4	1	3

Figure 4. State x of the 4×4 Sudoku.

- a) Explain what *CSP* is in your own words. (3 marks)
- b) Explain the reasoning behind the name *Alldiff*. (1 mark)
- c) Provide **TWO (2)** other examples of global constraints. (2 marks)
- d) Design the Sudoku as a *CSP* by identifying its *variables*, the *domain* for each variable, and the *constraints* using the *Alldiff* constraint. (9 marks)
- e) To simplify implementation, suppose all *Alldiff* constraints are expanded into binary constraints, allowing the *AC-3 algorithm* (the most popular algorithm for enforcing *arc consistency*) to be applied directly. Explain how the value of B1 is determined by making following variables *arc-consistent* step by step.
 - (i) B4 (2 marks)
 - (ii) A3 (2 marks)
 - (iii) A2 (2 marks)
 - (iv) D1 (2 marks)
 - (v) B1 (2 marks)

[Total: 25 marks]